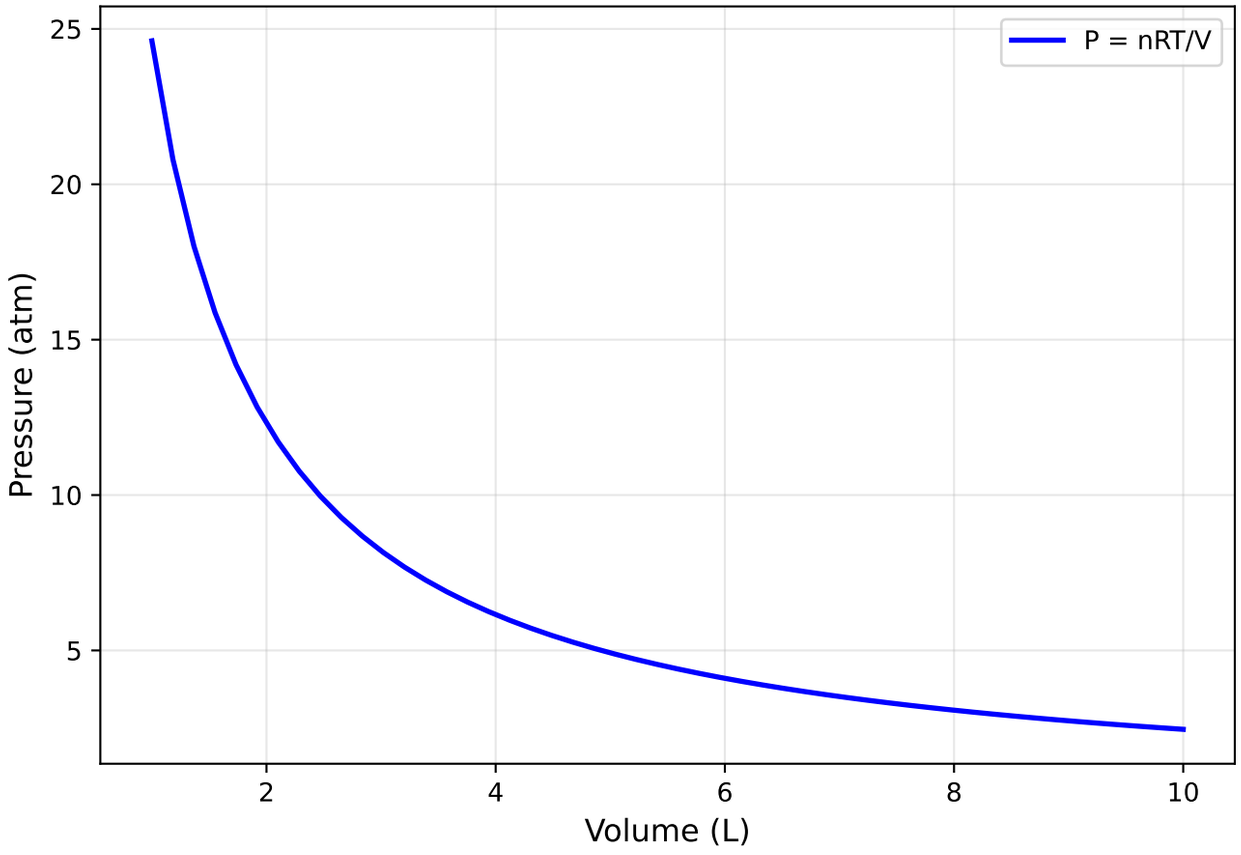
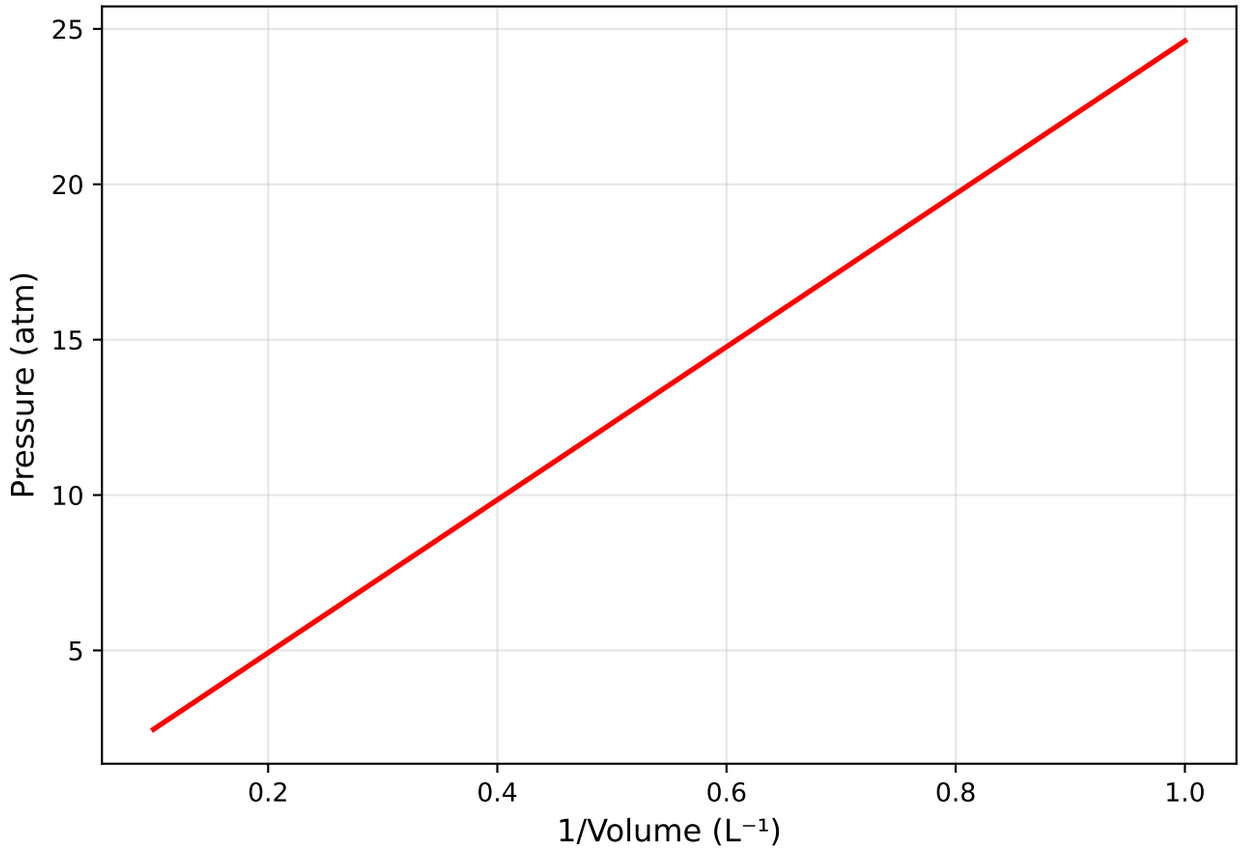
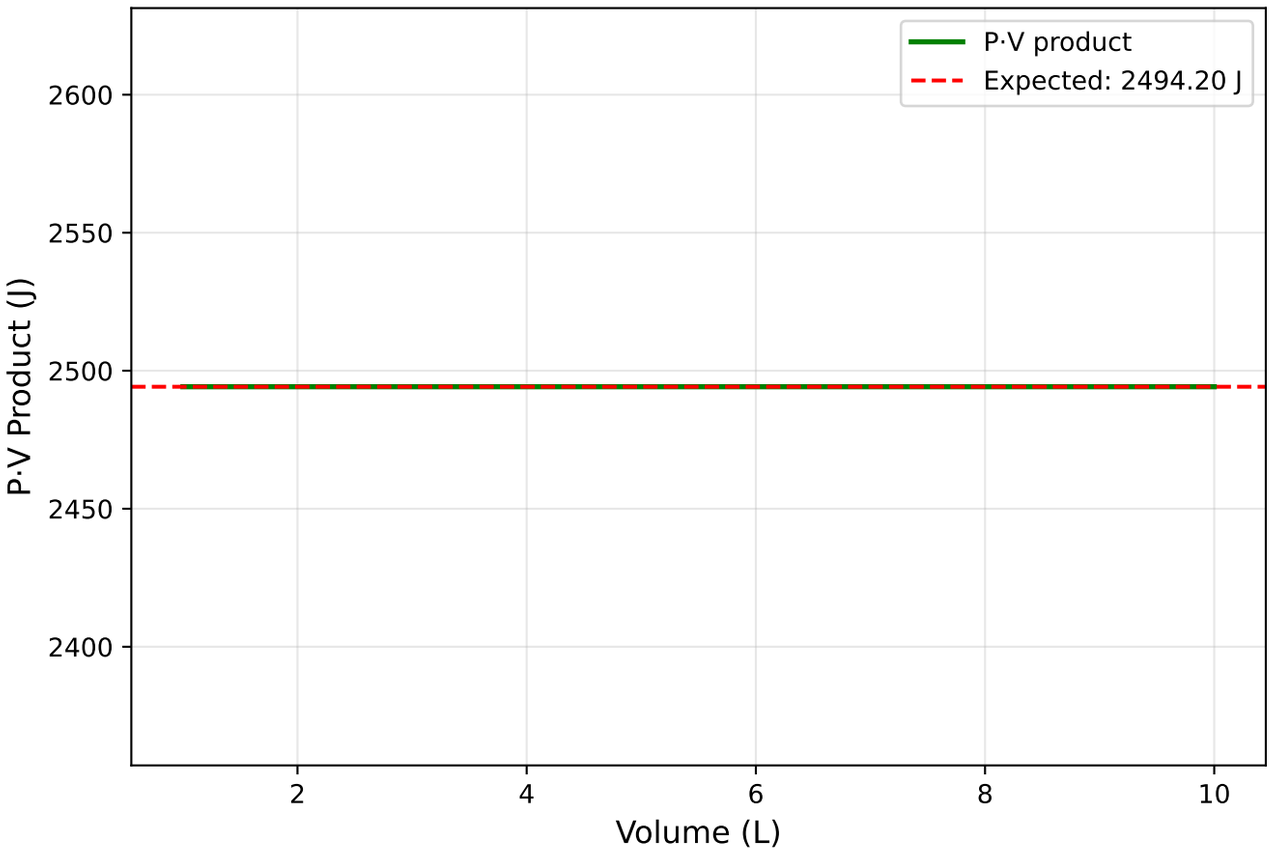
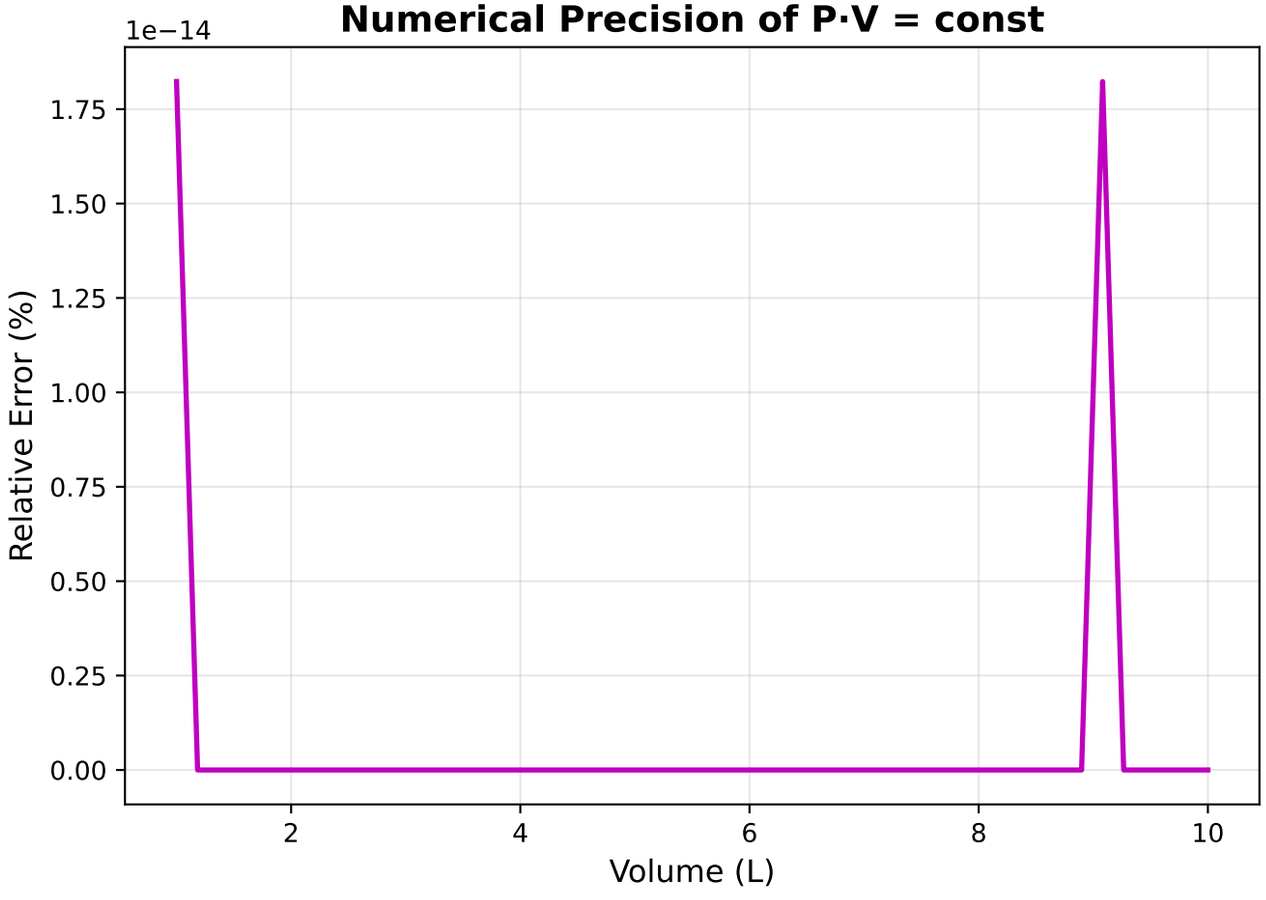

Boyle's Law: P vs V
P = nRT/V
Pressure (atm)
Volume (L)
25
20
15
10
5
2
4
6
8
10


Linear Verification: P ∝ 1/V
Pressure (atm)
1/Volume (L⁻¹)
25
20
15
10
5
0.2
0.4
0.6
0.8
1.0


Conservation of P·V Product
P·V product
Expected: 2494.20 J
P·V Product (J)
Volume (L)
2600
2550
2500
2450
2400
2
4
6
8
10


Numerical Precision of P·V = const
1e−14
Relative Error (%)
Volume (L)
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
2
4
6
8
10